Concavity of the Three-Variable Rademacher Moment throughout the Missing Range $2 < p < 3$

Run `fa_banach_001`, lane 17 (`agent_lane_17`)

13 August 2026

Status: new substantial partial result, likely valid, pending human review. For every $2 < p < 3$, the Schechtman concavity conjecture is proved for three Rademacher variables. Equivalently, the constant-one Hanner-type inequality holds for every three functions in real L_p. The unrestricted number of variables remains open.

1 Problem and exact scope

Zhang observes in Section 4, arXiv PDF page 9, that a proposed matrix concavity statement would imply the long-open assertion that the Rademacher sequence has concavity constant one [1, 2]. In his notation the exponent is $1/p$; we use the conventional moment exponent $p > 2$.

with $\alpha = ps$ and $\beta = qs$. By assumption, $0 < \alpha, \beta < 1$ and $\alpha + \beta \leqslant 1$. So (3.6) holds for $r = 2$ by Lemma 3.3. The inequality (3.6) for $r = \infty$ is a consequence of Russo–Dye Theorem (see [Bha09, Theorem 2.3.7] and [RD66]), since ϕ is unital, linear and positive. So the proof is finished. □

Remark 3.4. The argument here is similar to Beigi's proof of data processing inequality for certain sandwiched Rényi relative entropies [Bei13]. The endpoint case $r = 2$ here is non-trivial.

4. Non-concavity and non-convexity results

There are several reasons of expecting the concavity of

$$\Psi(A) := \mathrm{Tr}|K_1 A^p K_2|^{1/p}, \qquad A \in M_n^+(\mathbb{C}) \tag{4.1}$$

for $0 < p < 1/2$ and any $K_1, K_2 \in M_n^+(\mathbb{C})$. In [Che20] Chehade proposed a strategy to obtain equality conditions of data processing inequality for $\alpha - z$ Rényi relative entropies, where the concavity of (4.1) is the key. For more about the $\alpha - z$ Rényi relative entropies and their data processing inequalities, see [AD15, Zha20] and references therein.

Another application of this concavity result (if it were true) is the concavity of

$$\mathbb{R}_+^n \ni (u_1, \dots, u_n) \mapsto |\sum_{j=1}^n u_j^p r_j|^{1/p} \tag{4.2}$$

for any $\{r_j\}_{j=1}^n \subset \mathbb{R}$ and any $n \geqslant 1$. To see this, just take $K_1 = \sum_{j=1}^n |1\rangle\langle j|$, $A = \sum_{j=1}^n u_j |j\rangle\langle j|$ and $K_2 = \sum_{j=1}^n r_j |j\rangle\langle 1|$. The concavity of (4.2) will imply the concavity of

$$\mathbb{R}_+^n \ni (u_1, \dots, u_n) \mapsto \int |\sum_{j=1}^n u_j^p r_j|^{1/p}.$$

Here r_j's are Rademacher functions. So one may prove that the $1/p$-concavity constant of the Rademacher sequence in $L_{1/p}$ is 1, which was left open in [Sch95].

One evidence of supporting the concavity of (4.1) is the case when $K_1 = K_2^*$ (or $K_1 = \mathbf{1}$). In this case we know well the concavity of Ψ. See [Zha20] or the following Remark 4.1 for example. Unfortunately and surprisingly, for general K_1 and K_2, Ψ is not concave anymore.

More generally we have the non-concavity Theorem 1.3 as stated in the Introduction.

Proof of Theorem 1.3. We are not going to find any particular examples to disprove the concavity. The proof is based on some concavity and non-concavity results that are combined via the variational Lemma 2.1.

The matrix statement is disproved later in Zhang's paper, so that route does not answer the scalar problem. Jenkins and Tkocz subsequently isolate the precise remaining real range $2 < p < 3$ and the equivalent formulation below [3, Section 3.2].

Source problem. *For $2 < p < 3$, is*

$$\Phi_n(x_1, \dots, x_n) := \mathbb{E}\left|\sum_{k=1}^n x_k^{1/p} \varepsilon_k\right|^p \tag{1}$$

concave on $\mathbb{R}_+^n$ for every n, where the ε_k are independent Rademacher variables?

Schechtman's equivalences turn this into the constant-one inequality

$$\mathbb{E}\left\|\sum_{k=1}^n \varepsilon_k f_k\right\|_p^p \le \mathbb{E}\left|\sum_{k=1}^n \varepsilon_k \|f_k\|_p\right|^p \tag{2}$$

for arbitrary real L_p functions f_k [2, 3]. The two-variable case is classical Hanner. The theorem below settles the next case $n = 3$.

2 Proof intuition

The Hessian formula of Schechtman, in the convenient form recorded by Jenkins–Tkocz, writes minus the Hessian as the Laplacian of a weighted complete graph. For three variables this graph is a triangle. After the three coefficients are ordered, two edges have nonnegative weight, while only the edge joining the two smaller coefficients can be negative.

A weighted triangle with positive weights a, b and a negative weight $-c$ has a positive semidefinite Laplacian exactly when the bad conductance c is at most the effective conductance $ab/(a+b)$ of the other two edges. The required estimate is elementary but not termwise: in the genuine triangle regime it follows from two secant-ratio inequalities for t^q, $0 < q < 1$. Both reduce to the fact that

$$g(a, h) = a\big((a+h)^q - a^q\big)$$

is increasing in each variable. Thus the elusive "coarser grouping" of the Hessian terms becomes exact effective-resistance grouping when $n = 3$.

3 Laplacian reduction

We begin with an elementary criterion.

Lemma 1 (weighted triangle)**.** *Let $a, b > 0$ and $c \geq 0$. The quadratic form*

$$Q(z) = a(z_1 - z_2)^2 + b(z_1 - z_3)^2 - c(z_2 - z_3)^2$$

is nonnegative on $\mathbb{R}^3$ if and only if

$$c(a+b) \leq ab. \tag{3}$$

Proof. For a fixed value of $z_2 - z_3$, minimize the first two terms over z_1. Their minimum is $ab(z_2 - z_3)^2/(a+b)$. This gives (3). $\square$

For $x_k > 0$, put $r_k = x_k^{1/p}$ and $S = \sum r_k \varepsilon_k$. The Hessian identity from [3, Lemma 3] says that for every $h \in \mathbb{R}^n$,

$$h^\mathsf{T} D^2 \Phi_n(x) h = -\frac{p-1}{p} \sum_{k<\ell} \left(\frac{h_k}{x_k} - \frac{h_\ell}{x_\ell} \right)^2 r_k r_\ell \mathbb{E}\big[|S|^{p-2} \varepsilon_k \varepsilon_\ell\big]. \tag{4}$$

Thus Φ_n is concave exactly when the associated weighted Laplacian is positive semidefinite.

4 Main theorem

Theorem 1. *Let $2 < p < 3$ and let $\varepsilon_1, \varepsilon_2, \varepsilon_3$ be independent Rademacher variables. Then*

$$(x_1, x_2, x_3) \longmapsto \mathbb{E}\left| x_1^{1/p}\varepsilon_1 + x_2^{1/p}\varepsilon_2 + x_3^{1/p}\varepsilon_3 \right|^p$$

is concave on $\mathbb{R}^3_+$.

Proof. It suffices first to work on the positive orthant. Put $q = p - 2 \in (0, 1)$ and, after permuting indices, write the three root coefficients as

$$r \geq u \geq v > 0.$$

Define

$$\begin{array}{llll} A = r+u+v, & B = r+u-v, & C = r-u+v, & D = |r-u-v|, \\ U = A^q, & V = B^q, & W = C^q, & X = D^q, \end{array}$$

and

$$P = U+V-W-X, \qquad Q = U-V+W-X, \qquad R = U-V-W+X. \tag{5}$$

Direct averaging over the eight signs gives the three weights in (4):

$$w_{12} = \frac{ru}{4}P, \qquad w_{13} = \frac{rv}{4}Q, \qquad w_{23} = \frac{uv}{4}R. \tag{6}$$

Because $A > C$ and $B \geq D$, the decomposition $P = (U-W)+(V-X)$ gives $P > 0$. Similarly, $A > B$ and $C \geq D$ give $Q = (U-V)+(W-X) > 0$. If $R \geq 0$, all three weights are nonnegative and there is nothing to prove. Hence assume $R < 0$ and put

$$m = -R = -U+V+W-X > 0.$$

Lemma 1 and (6) reduce positive semidefiniteness to

$$m(uP+vQ) \leq rPQ. \tag{7}$$

First suppose $r \geq u+v$. From (5),

$$P-m = 2(U-W) \geq 0, \qquad Q-m = 2(U-V) \geq 0.$$

Therefore

$$m(uP+vQ) \leq uPQ+vPQ \leq rPQ,$$

which proves (7) in the dominant-coefficient cone.

It remains to consider the triangle regime $u \leq r < u+v$. Here $D = u+v-r$. We claim the sharper pair of estimates

$$\frac{m}{Q} \leq \frac{r-v}{u}, \qquad \frac{m}{P} \leq \frac{r-u}{v}. \tag{8}$$

Adding u times the first and v times the second gives exactly

$$m\left(\frac{u}{Q}+\frac{v}{P}\right) \leq (r-v)+(r-u) < r,$$

where the last inequality is $r < u+v$. Multiplication by PQ proves (7).

We prove (8). The first inequality is equivalent, after substituting (5), to

$$(r+u-v)(U-V) \geq (u+v-r)(W-X), \tag{9}$$

and the second to

$$(r-u+v)(U-W) \geq (u+v-r)(V-X). \tag{10}$$

For $a, h > 0$ set

$$g(a,h) = a\big((a+h)^q - a^q\big).$$

Clearly g is increasing in h. It is also increasing in a, because with $t = h/a$,

$$\frac{\partial g}{\partial a} = a^q\big((1+q)((1+t)^q-1) - qt(1+t)^{q-1}\big) = a^q\big((1+t)^{q-1}(1+q+t)-(1+q)\big) > 0.$$

Indeed $F(t) = (1+t)^{q-1}(1+q+t)$ satisfies $F(0) = 1+q$ and $F'(t) = q(1+t)^{q-2}(q+t) > 0$.

Now (9) is

$$g(B, 2v) \geq g(D, 2(r-u)),$$

which follows from $B \geq D$ and $v \geq r - u$. Likewise (10) is

$$g(C, 2u) \geq g(D, 2(r - v)),$$

which follows from $C \geq D$ and $u \geq r - v$. This proves (8), hence (7), and therefore the Hessian is negative semidefinite throughout $(0, \infty)^3$.

Finally, Φ_3 is continuous on the closed positive orthant. Concavity on the open orthant extends to the boundary by approximation, completing the proof. $\square$

Corollary 1. *For $2 < p < 3$ and arbitrary real $f_1, f_2, f_3 \in L_p$,*

$$\mathbb{E}\, \|\varepsilon_1 f_1 + \varepsilon_2 f_2 + \varepsilon_3 f_3\|_p^p \leq \mathbb{E}\, |\varepsilon_1 \|f_1\|_p + \varepsilon_2 \|f_2\|_p + \varepsilon_3 \|f_3\|_p|^p .$$

Thus the three-dimensional Rademacher subspace has p-concavity constant one.

Proof. This is the $n = 3$ instance of Schechtman's equivalence between concavity of (1) and (2); see [2, 3]. $\square$

5 Eight focused upgrade attempts

After the live problem was identified, eight materially different or deepening routes were pursued.

1. **Exact-status audit.** Zhang's stronger matrix conjecture is refuted in the source itself. Jenkins–Tkocz confirms that the scalar real problem remains open precisely for $2 < p < 3$.
2. **Hessian/Laplacian reduction.** Formula (4) turns the sign problem into positive semidefiniteness of a signed graph Laplacian. This is the successful structural reduction used above.
3. **Counterexample search.** Exact sign enumeration and eigenvalue tests over broad logarithmic coefficient samples for $3 \leq n \leq 9$ found no contradiction, while confirming that individual edge weights can be negative.
4. **Bernstein integral representation.** Representing $|S|^q$ by a Bernstein integral does not prove positivity pointwise: sampled integrand Laplacians can be indefinite. Integration must exploit cancellation.
5. **Dominant three-variable cone.** The effective-conductance criterion and the simple inequalities $m \leq P, Q$ prove the case $r \geq u + v$ directly.
6. **Triangle three-variable cone.** The sharp ratio bounds (8), proved by monotonicity of $g(a, h)$, complete the new $n = 3$ theorem.
7. **Interpolation from $p = 2$ and $p = 3$.** The Hessian depends on both the power $q = p - 2$ and the nonlinear coordinates $x_k^{1/p}$; neither matrix convexity in p nor a suitable complex interpolation family was found. Endpoint positivity alone does not control the interior signed edges.
8. **Extension by triangle decomposition.** A tempting upgrade is to decompose the n-vertex Laplacian into positive three-vertex Laplacians. This fails termwise: after conditioning on the other signs, a three-vertex sub-Laplacian may have all three edge weights negative. A genuinely global grouping remains necessary for $n \geq 4$.

6 Verification, novelty, and limitations

The formal proof uses only the cited Hessian identity, the elementary weighted-triangle lemma, and explicit inequalities for t^q. The included script `code/check_hessian.py` independently enumerates all eight signs, checks (6), and tests the Laplacian over 200,000 fixed random samples spanning eight exponents and twelve logarithmic orders of magnitude. Its worst scaled minimum eigenvalue is $-4.24\cdot 10^{-16}$ (roundoff), and the maximum relative formula discrepancy is $1.44\cdot 10^{-13}$. This computation is a contradiction check, not part of the proof.

A bounded novelty search covered the run's four cheap indexes, the source paper, Schechtman's cited problem, Jenkins–Tkocz and its references, and focused arXiv/web queries combining "Rademacher", "p-concavity", "$2 < p < 3$", "three variables", and Hanner's inequality. No prior three-variable statement or proof was found. The search is not exhaustive, so novelty confidence is moderate rather than definitive.

The result does not settle arbitrary n. Human review should prioritize: the sign claims $P, Q \geq 0$; the algebra converting (8) to (9)–(10); monotonicity of g; and the precise use of Schechtman's equivalence in Corollary 1.

References

[1] H. Zhang, *Some convexity and monotonicity results of trace functionals*, J. Math. Phys. **63** (2022), 091901; arXiv:2108.05785.

[2] G. Schechtman, *Two remarks on 1-unconditional basic sequences in L_p,* $3 \leq p < \infty$, in *Geometric Aspects of Functional Analysis*, Oper. Theory Adv. Appl. 77, Birkhäuser, 1995, 251–254.

[3] J. Jenkins and T. Tkocz, *Complex Hanner's inequality for many functions*, Proc. Amer. Math. Soc. **151** (2023), 1191–1202; arXiv:2207.09122.